

# DoangJoo "Alan" Synn

PhD Candidate in Computer Science · Georgia Institute of Technology

Advised by [Prof. HyunJoo Oh](#) and [Prof. Sehoon Ha](#)

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## Research Interests

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I study **motion** as a computational medium across HCI and computer graphics. My research builds systems for designing, generating, and fabricating motion. I aim to make motion programmable and tangible across animated characters, mechanical artifacts, robotic systems, and AI-assisted creative tools.

Before starting my Ph.D., I spent eight years building software systems across AI systems, games, cloud platforms, and social robots.

## Education

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### Georgia Tech, Ph.D. in Computer Science

2022 – Present

- Atlanta, GA, USA
- [School of Interactive Computing](#); advised by Profs. [Oh](#) & [Ha](#)
- Research: computational design, mechanical crafting, and motion synthesis (HCI/Graphics)

### Korea University, M.Eng. in ECE

2020 – 2022

- Seoul, Korea
- Advised by [Prof. Jong-Kook Kim](#)
- GPA: 4.25 (4.5 scale)
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

### Korea University, B.Eng. in EE

2012 – 2020

- Seoul, Korea
- GPA: 3.56 (4.5 scale)
- Military service leave as a Software Engineer, 2016–2019

## Professional Experience

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### PhD Research Intern, Dolby Laboratories

2024 – 2025

- Atlanta, GA, USA
- Experience Delivery Lab, built a time-series, multimodal datastore supporting low-latency media analytics.

### MLOps Engineering Intern, LG Uplus Corp.

2023

- Seoul, Korea
- Led **DeepCrunch**, a model-compression library with multi-backend AI inference for efficient edge serving.

### Research Scientist, Protopia AI Inc.

2021 – 2022

- Austin, TX, USA

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- CEO: [Dr. Ebrahimi](#); CTO: [Prof. Esmailzadeh](#).

#### Cloud Architect · [AWS Educate Cloud Ambassador](#)

2020 – 2022

- Seoul, Korea
- Led engineering for 4 startup services, translating ambiguous product needs into back-end systems while mentoring early-career engineers.
- Built production cloud infrastructure with Kubernetes, Terraform, CI/CD, and cost-optimized autoscaling.

#### Server Engineer, Flysher Inc.

2019

- Seoul, Korea
- Shipped back-end features for 4 social-game titles; one reached global top-10 on Facebook.
- Led an ETL → Redshift analytics pipeline driving data-informed feature rollout.

#### System Engineer, [AKA Intelligence \(Musio\)](#)

2017 – 2019

- Seoul, Korea
- Designed embedded Linux boards & firmware for the social robot **Musio**.
- Coordinated HW-SW integration and production QA with overseas ODMs.

## Publications

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### Refereed Journal Articles

- 1 M. Ryu, X. Piao, J. Park, **D. Synn**, and J. Kim, “ADaPT: An Automated Dataloader Parameter Tuning Framework using AVL Tree-based Search Algorithms,” *TKIPS*, Jan. 2025.
- 2 **D. Synn**, X. Piao, J. Park, and J. Kim, “Distributed Training of Deep Learning Models Using Micro Batch Streaming,” *KTSDE*, Mar. 2023.
- 3 X. Piao, **D. Synn**, J. Park, and J. Kim, “Enabling large batch size training for dnn models beyond the memory limit while maintaining performance,” *IEEE Access*, 2023.
- 4 J. Han, **D. Synn**, T. Kim, H. Jung, and J. Kim, “Feature Based Sampling: A Fast and Robust Sampling Method for Tasks Using 3D Point Cloud,” *IEEE Access*, Apr. 2022.

### Refereed Conference Papers

- 1 **D. Synn**, Z. Guo, S. Ha, and H. Oh, “MotionSmith: A Sketch-Based Design System for Automata Making,” *CHI*, Apr. 2026.
- 2 J. Yu, P. Garg, **D. Synn**, and H. Oh, “Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning,” *CHI*, Apr. 2025.
- 3 **D. Synn** and J. Kim, “Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders,” *ACK*, Oct. 2021.
- 4 J. Park, **D. Synn**, and J. Kim, “A Survey on the Advancement of Virtualization Technology,” *KIPS*, May 2021.
- 5 B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmailzadeh, “Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection,” *NeurIPS*, 2021.
- 6 D. Kim, **D. Synn**, and J. Kim, “SG-Drop: Faster Skip-Gram by Dropping Context Words,” *KIPS*, Nov. 2020.

### Preprints

- 1 J. Park, **D. Synn**, X. Piao, and J. Kim, “Dataloader Parameter Tuner: An Automated Dataloader Parameter Tuner for Deep Learning Models,” *arXiv*, 2022.

## Honors & Awards

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### Fellowships

- [Brain Korea 21 \(BK21\)](#) Research Fellowship (2020 - 2022)

### Awards

- Dean's List, Spring 2012 & Fall 2020, Electrical Engineering
- Prize Winner, Creative Challenge Program @ [Korea Univ. CTL](#) (2016)
- Prize Winner, Creative Challenge Program @ [Korea Univ. CTL](#) (2015)

## Teaching

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### Head TA, CS 4496/7496 Computer Animation

2024 – Present

- [Georgia Tech](#), [School of Interactive Computing](#).

### Lecturer, Korea University

2021 – 2022

- `Java101` for 10+ undergraduate students.

### Teaching Assistant, Korea University

2021

- Digital System Design
- Data Structure and Algorithm

### Lecturer, KAIST Engineering School

2012 – 2014

- Co-developed national coding education programs with South Korea's Ministry of Science and ICT.
- Designed 20+ project-based curricula combining Arduino, hardware prototyping, and software development.
- Evaluated and iterated on newly developed curricula through pilot programs with high-school and college students.

## Activities & Service

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### Georgia Tech Korean Student Association

2023 – 2024

- President of Korean Students in the College of Computing.

### Hardware & Software Community, Korea University

2014–2019

- President of Korea University's hardware and software engineering community.
- Grew the community from 10 members to 200+ active participants across hardware and software tracks.

### TEDxKoreaUniversity, Korea University

2015 – 2016

- Founder & Chief Organizer.
- Organized a TEDx event with 200 guests, 7 speakers, 47 team members, and a \$4k budget. Responsibilities included fundraising, sponsorships, editorial decisions, communication planning, and production & media.

## References

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**Professor HyunJoo Oh**

Assistant Professor

Georgia Institute of Technology

School of Industrial Design & School of Interactive Computing

**Professor Sehoon Ha**

Assistant Professor

Georgia Institute of Technology

College of Computing, School of Interactive Computing

**Dr. ByungHoon Ahn**

Senior Software Engineer, Apple Inc.

[bhahn@ucsd.edu](mailto:bhahn@ucsd.edu)

**Dr. Eiman Ebrahimi**

CEO, Protopia AI Inc.

<http://eimanebrahimi.com>

**Professor Hadi Esmailzadeh**

CTO, Protopia AI Inc.

Associate Professor, University of California San Diego

<http://cseweb.ucsd.edu/~hadi/>